Week 1 Level 1: Introduction

Robots Need Instructions

A program is a list of exact steps, done in order.

Our class agreements

Digital citizenship & safety: How we work together

1. **Be kind.** We laugh with each other, never at a mistake. Bugs are normal!
2. **Share the mouse.** The driver clicks, the navigator helps. We swap when the timer rings.
3. **Point, don't grab.** To help a partner, point at the screen and explain.
4. **Nicknames only.** No real full names, addresses, school names, birthdays or photos in our projects.
5. **Offline and private.** Scratch stays on this Mac. No accounts, no sharing online. If something asks for your name or a password, stop and get a grown-up.
6. **Look after your body.** Take stretch breaks and rest your eyes.

My coding nickname: _____ I agree: _____

What you will learn

- What an **algorithm** is, and why the order of steps matters.
- How to find your way around the Scratch app.
- How to snap blocks together and start them with the **green flag**.
- How to find and fix a bug when steps are in the wrong order.

The big idea

A computer is like a robot. It does **exactly** what you tell it, **in the order** you tell it. It never guesses what you meant.

A list of exact steps in order is called an **algorithm**. When a computer follows an algorithm, we call it a **program**. A mistake in a program is a **bug**. Finding and fixing it is called **debugging**.

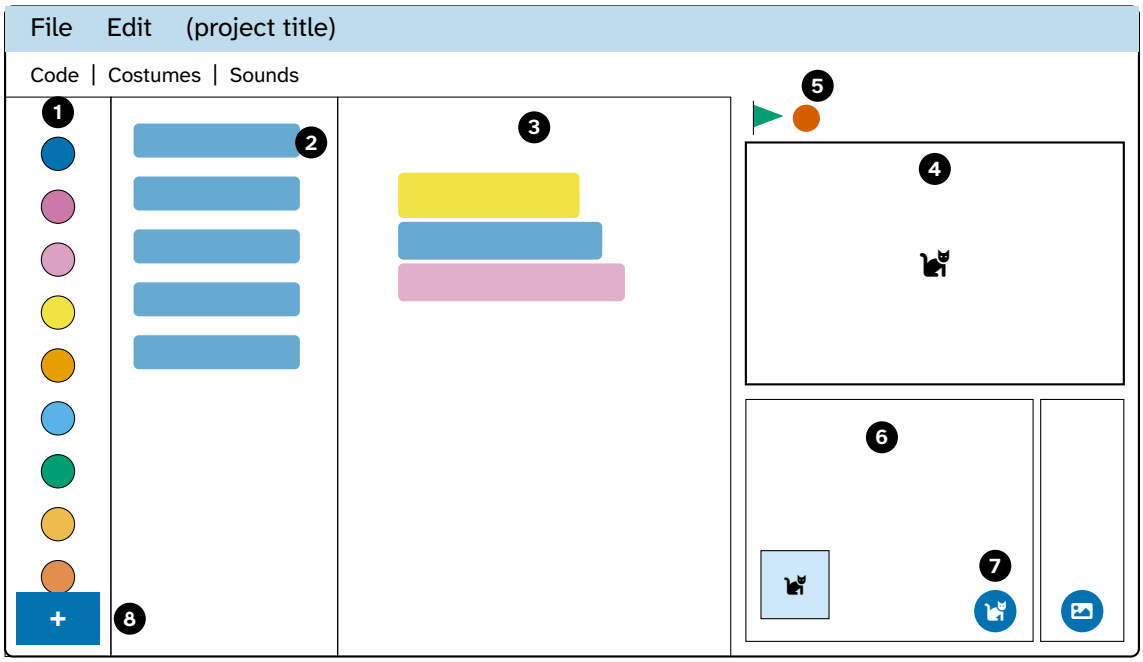
Tip: Try it on paper

Number these steps 1 to 5 so a robot can make a jam sandwich.

_____ Spread jam on one slice.
_____ Put the other slice on top.
_____ Take two slices of bread.
_____ Open the jam jar.
_____ Eat and enjoy!

Meet Scratch

Scratch is a coding app. You make programs by snapping blocks together, like puzzle pieces.



1 Block categories (colored circles: Motion, Looks, Sound, Events, ...)
2 Blocks you can drag
3 Code area: snap blocks here to make a **script**
4 Stage: where your sprites act
5 Green flag (start) and red stop sign
6 Sprite pane: all your sprites
7 Choose a Sprite button (and Choose a Backdrop, far right)
8 Add Extension button (we use it in week 2)

Exercise 1.1: Hello, Cat!

Make the cat say hello, walk and say bye.

1. **Events**: drag “when green flag clicked” into the code area.

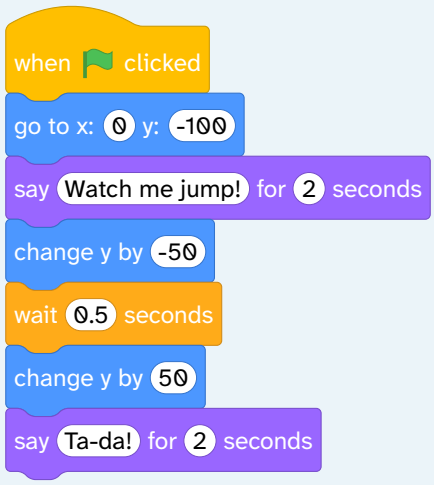
2. **Motion**: snap “go to x: 0 y: 0” under it. That is the middle of the stage.
3. **Looks**: snap “say Hello! for 2 seconds”. Type “Hi! I am” and your nickname.
4. **Motion**: snap “move 10 steps”. Change 10 to 100.
5. **Sound**: snap “play sound Meow until done”.
6. **Looks**: snap one more “say … for 2 seconds” and type “Bye!”.
7. Click the green flag. Save: **File** → **Save to your computer**, folder `Scratch Projects`.



Done when: the cat talks, walks, meows and says bye, in that order, every time you click the flag.

Exercise 1.2: Mixed-Up Jump (bug hunt)

The cat should say “Watch me jump!”, jump **up**, come back down to the ground and say “Ta-da!”. Build this script **exactly** as it is, bug and all. (**change y by** is in **Motion**: y is up and down.)



Click the flag. What did the cat do?

Which two blocks are in the wrong order?

Now drag them into the right order and test again. Did the cat jump up? ☐ yes

Tip: Debugging steps

1. What did I **expect**? 2. What **happened**? 3. Read the blocks one at a time with your finger. 4. Change **one** thing and test again.

Exercise 1.3: My Sprite Show

Make your own little show.

1. Click **Choose a Sprite** (the cat-face button, bottom right) and pick any sprite you like.
2. Click **Choose a Backdrop** (far bottom right) and pick a place for your show.
3. Plan 5 steps below, then build them under “when green flag clicked”.

My show plan (one step per line):

Done when: your show starts with the green flag and the steps happen in your planned order.

Vocabulary box

algorithm A list of exact steps, in order, that solves a problem.

program An algorithm that a computer can follow.

sequence Steps that happen one after another, in order.

bug / debugging A mistake in a program / finding and fixing it.

sprite A character or object on the Scratch stage, like the cat.

stage The area where your sprites act.

script A stack of blocks snapped together.

green flag The button that starts your program.

driver / navigator The driver clicks; the navigator reads, checks and helps.

🖌 Reflection

What surprised you about giving instructions to a robot?

One thing I want to make in Scratch:

Week 1
Robots Need Instructions
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15–20 minutes. Do Core first; Challenge and Creative are optional.

Name (nickname is fine): _____ Date: _____

★ Core (everyone): Walk and Talk (Scratch, about 10 minutes)

Make a sprite walk across the stage and talk, in the right order. (No Mac at home today? Do the Challenge on paper instead and build this next week.)

1. Open the Scratch app. Click **Choose a Backdrop** (bottom right) and pick any place.
2. Build this script for the cat (or any sprite you choose). **glide** is in **Motion**.
3. Click the green flag. The sprite should start on the left, talk, glide to the middle, talk, glide to the right and say bye.
4. Save it: **File** → **Save to your computer**, name it `walk-and-talk`.



Use your own nickname and your own words in the speech bubbles.

★★ Challenge (want more?): Be the programmer for your family robot

Write an algorithm for brushing your teeth in exactly 6 steps. Then ask someone at home to be the robot and follow your steps **exactly**, without guessing. Did they do it right? If not, fix your steps.

What did you have to fix?

Creative (make it yours): A three-scene story

Add a “switch backdrop to …” block (in **Looks**) between the glides so the sprite walks into a new place each time. Choose a backdrop for each part of your story.

Self-check: tick each box when it is true

- ☐ My script starts with “when green flag clicked”.
- ☐ My sprite always starts on the left, because of the “go to x: -180 y: -100” block.
- ☐ The steps happen in the order I planned.
- ☐ I used my nickname, not my real name.
- ☐ I saved my project.